

CUSTOMER BEHAVIOUR ANALYSIS

Northwind Database | SQL Server

OVERVIEW

Tool Used: SQL Server (SSMS)

Dataset: Northwind Database

Records Analyzed: 91 Customers | 830 Orders | 2,155 Order Details

Categories: 8 | Employees: 9 | Countries: 21

PROBLEM:

A retail business needed to understand which customers, products, regions and employees drive the most revenue, and where operational improvements could be made to increase profitability.

DATA

The Northwind database contains transactional data from a global retail company including customer orders, product details, employee records and shipping information across 21 countries from 1996–1998.

MY PROCESS

- Restored and explored Northwind database across 6 related tables
- Wrote 10 SQL queries of increasing complexity in SQL Server
- Used JOINS across 3 tables, GROUP BY aggregations, DATEDIFF
- date functions, TOP N filtering, and a CTE with subquery
- Identified revenue drivers across customers, products,
- categories, countries and employees

KEY FINDINGS

1. Total business revenue stands at \$1,265,793.04 across all orders.
2. Quick-Stop (Germany) is the top customer generating \$110,277.30, the highest of any single customer.

```
1460 -- Q18: High value customers above average spend using a CTE
1461 WITH CustomerRevenue AS (
1462     SELECT
1463         C.CompanyName,
1464         C.Country,
1465         ROUND(SUM(OD.UnitPrice * OD.Quantity * (1 - OD.Discount)), 2) AS Total_Revenue
1466     FROM Customers C
1467     JOIN Orders O ON C.CustomerID = O.CustomerID
1468     JOIN [Order Details] OD ON O.OrderID = OD.OrderID
1469     GROUP BY C.CompanyName, C.Country
1470 )
1471 SELECT
1472     CompanyName,
1473     Country,
1474     Total_Revenue
1475 FROM CustomerRevenue
1476 WHERE Total_Revenue > (SELECT AVG(Total_Revenue) FROM CustomerRevenue)
1477 ORDER BY Total_Revenue DESC;
1478 -- FINDING: 30 out of 91 customers (33%) are classified as high value,
1479 -- spending above the average customer revenue. Quick-Stop remains the
1480 -- top customer confirming their importance to the business.
1481 -- These 30 customers should be the focus of retention strategies.
```

Ln: 168, Ch: 1 | SPC | CRLF | Windows 1

	CompanyName	Country	Total_Revenue
1	QUICK-Stop	Germany	110277.3
2	Ernst Handel	Austria	104874.98
3	Save-a-lot M.	USA	104361.95
4	Rattlesnake	USA	51097.8
5	Hungry Owl A.	Ireland	49979.91
6	Hanari Carnes	Brazil	32841.37
7	Königlich Ess.	Germany	30908.38
8	Folk och fa HB	Sweden	29567.56
9	Mère Pailarde	Canada	28872.19
10	White Clover	USA	27363.61
11	Frankenvers	Germany	26556.56
12	Queen Cozinha	Brazil	25717.5
13	Berglunds sn	Sweden	24927.58

3. Beverages is the highest revenue category while Grains/Cereals performs the lowest across all 8 categories.

4. Camembert Pierrot is the best-selling product by volume (1,577 units) however Tarte au sucre generates the most product revenue at \$47,234.97, showing that volume does not equal profit.

5. Q4 (October–December) is consistently the peak trading period across all years of available data.

6. The USA is the most valuable market leading in both customer count and total revenue generation.

```
99  -- Q7: Revenue by customer country
100  SELECT
101  C.Country,
102  COUNT(DISTINCT O.OrderID) AS Total_Orders,
103  ROUND(SUM(OD.UnitPrice * OD.Quantity * (1 - OD.Discount)), 2) AS Total_Revenue
104  FROM Customers C
105  JOIN Orders O ON C.CustomerID = O.CustomerID
106  JOIN [Order Details] OD ON O.OrderID = OD.OrderID
107  GROUP BY C.Country
108  ORDER BY Total_Revenue DESC;
109  -- FINDING: USA generates the most revenue and also has the most customers,
110  -- confirming it is the most valuable market both in volume and spend.
111
```

	Country	Total_Orders	Total_Revenue
1	USA	122	245584.61
2	Germany	122	230284.63
3	Austria	40	128003.84
4	Brazil	83	106925.78
5	France	77	81358.32
6	UK	56	58971.31
7	Venez...	46	56810.63
8	Sweden	37	54495.14
9	Canada	30	50196.29
10	Ireland	19	49979.91
11	Belgium	19	33824.85
12	Denmark	18	32661.02
13	Switzer...	18	31692.66
14	Mexico	28	23582.08
15	Finland	22	18810.05
16	Spain	23	17983.2
17	Italy	28	15770.15
18	Portugal	13	11472.36
19	Argent...	16	8119.1
20	Norway	6	5735.15
21	Poland	7	3531.95

7. Margaret Peacock is the top performing employee handling 156 orders and \$232,890.85 in revenue — 18% of total business revenue.

129	
130	
131	-- Q9: Employee performance by orders and revenue
132	SELECT
133	E.FirstName + ' ' + E.LastName AS Employee_Name,
134	COUNT(DISTINCT O.OrderID) AS Total_Orders,
135	ROUND(SUM(OD.UnitPrice * OD.Quantity * (1 - OD.Discount)), 2) AS Total_Revenue
136	FROM Employees E
137	JOIN Orders O ON E.EmployeeID = O.EmployeeID
138	JOIN [Order Details] OD ON O.OrderID = OD.OrderID
139	GROUP BY E.FirstName, E.LastName
140	ORDER BY Total_Revenue DESC;
141	-- FINDING: Margaret Peacock is the top performing employee handling
142	-- 156 orders and generating \$232,890.85 in revenue, leading both
143	-- in order volume and total revenue generated.
144	
145	

Employee_Name	Total_Orders	Total_Revenue
1	Margaret Peacock	156
2	Janet Leverling	127
3	Nancy Davolio	123
4	Andrew Fuller	96
5	Laura Callahan	104
6	Robert King	72
7	Anne Dodsworth	43
8	Michael Suyama	67
9	Steven Buchanan	42

8. Federal Shipping delivers orders fastest at 7.5 days average

vs United Package at 9.2 days.

9. Only 30 of 91 customers (33%) spend above average, making

high value customer retention a critical business priority.

RECOMMENDATION

The business should prioritise retention of its 30 high-value customers who drive the majority of revenue, with Quick-Stop and Ernst Handel requiring dedicated account management. Shifting more orders to Federal Shipping would improve delivery times and customer satisfaction. The Grains/Cereals category should be reviewed for consolidation or replacement with higher performing product lines. Finally, Q4 stock and staffing levels should be increased to capitalise on the consistently high seasonal demand.

SQL TECHNIQUES USED

- SELECT, WHERE, ORDER BY
- GROUP BY with SUM, COUNT, AVG, ROUND
- JOINS across multiple tables
- TOP N filtering
- DATEDIFF and DATENAME date functions
- CAST for data type conversion
- CTE (Common Table Expression)
- Subquery within WHERE clause

